Lily's PAD
3D Exploded Assembly View

LILY'S PAD

MODULE ELEC 06/26 - 10/26

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Overview

Lily's Pad is a simple interactive and educational object primarily intended for children (5+), teachers/educational teams, and families

It allows users to familiarize themselves with the alphabet and numbers in French Sign Language (FSL) or British Sign Language (BSL), as chosen.

Lily's Pad is also accessible for hearing users who are curious or who have hearing-impaired loved ones they would like to communicate with.

Because we are aiming for a market of young children mainly, great care has been taken regarding the presentation of the object. To make it as attractive as possible. So we have designed a casing in the form of a frog upon a lily pad. (cf. image above)

Video promotionnelle

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I. Features

1. An LCD screen connected to a keyboard featuring letters and numbers.
2. A rotary encoder allowing users to scroll through options on the screen, it is clickable to confirm an input.
3. A USB Type-C port to power the device.
4. A POWER switch to turn the device on/off.
5. A RESET button to reboot the device.
6. A 3D-printed casing around the screen.
7. A 3D-printed base in which the keyboard is embedded.
8. A slide switch to change language, English (BSL) or French (FSL).
9. Two small LCD screens to display animated eyes.
10. Translation/game/alphabet modes.

II. User Manual

The Lily's Pad is a device composed of a main screen, a keyboard, a rotary encoder for navigating through the software and a slide switch to select the device's language.

The device, taking the shape of a frog sitting on a lily pad, features two round screens serving as animated eyes.

To operate, the device must be plugged in via USB Type-C, then simply use the POWER switch to turn it on or off.

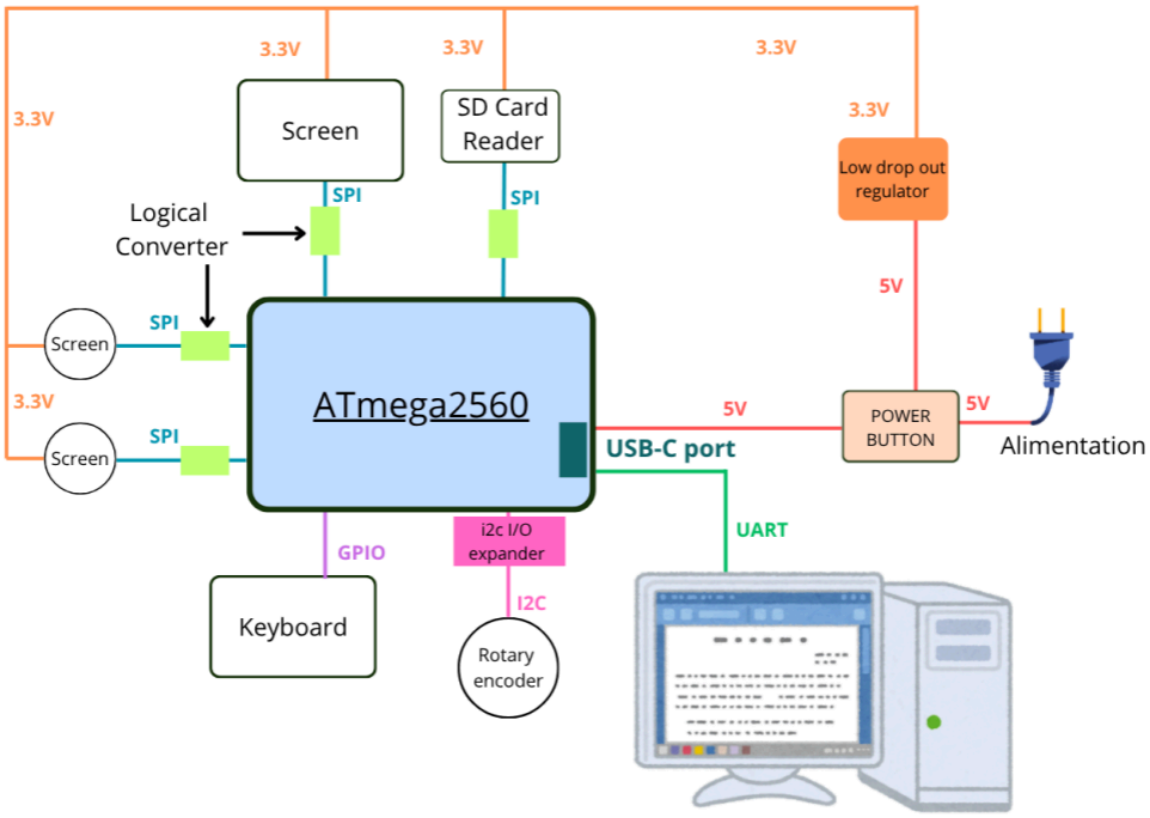
A home screen appears with the option to choose from the following modes:

Translation	The user enters a letter or a word (letter by letter) from the keyboard then it is spelled on the LCD screen
Game	There are two versions of the game: - The screen displays a letter, and the user must select the corresponding sign-language image using the rotary encoder. - The screen displays a sign-language image, and the user must choose the corresponding letter.
Alphabet	To display the sign-language alphabet, the user rotates the encoder to scroll through the corresponding hand-sign images.

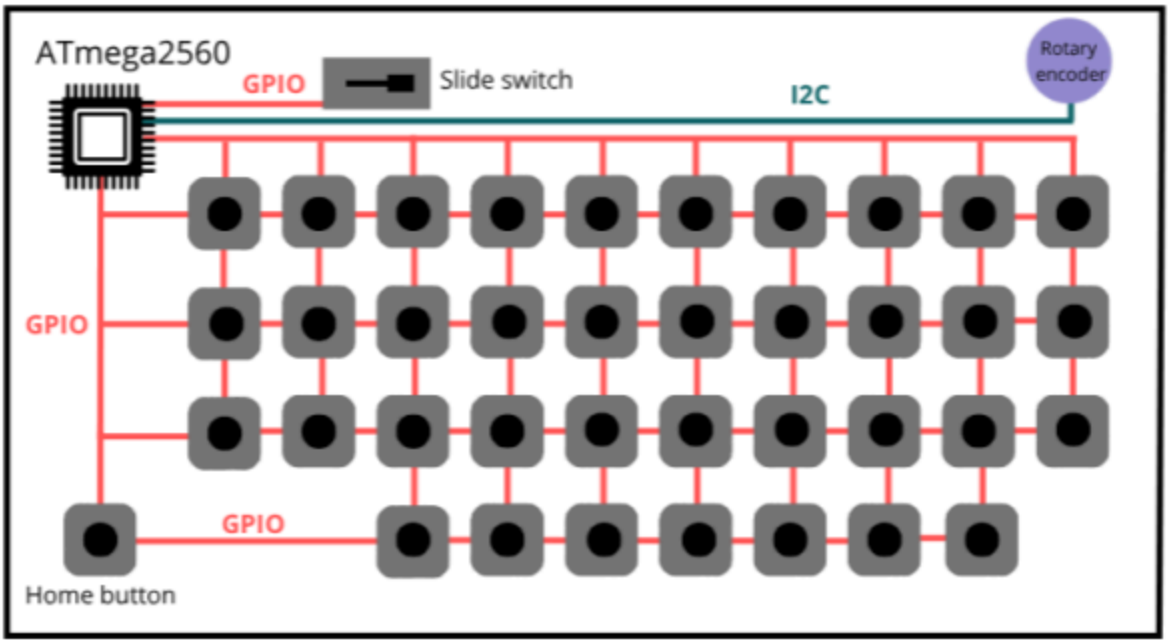
The slide switch is used to select whether to use BSL or LSF because the signs are different.

III. Diagrams

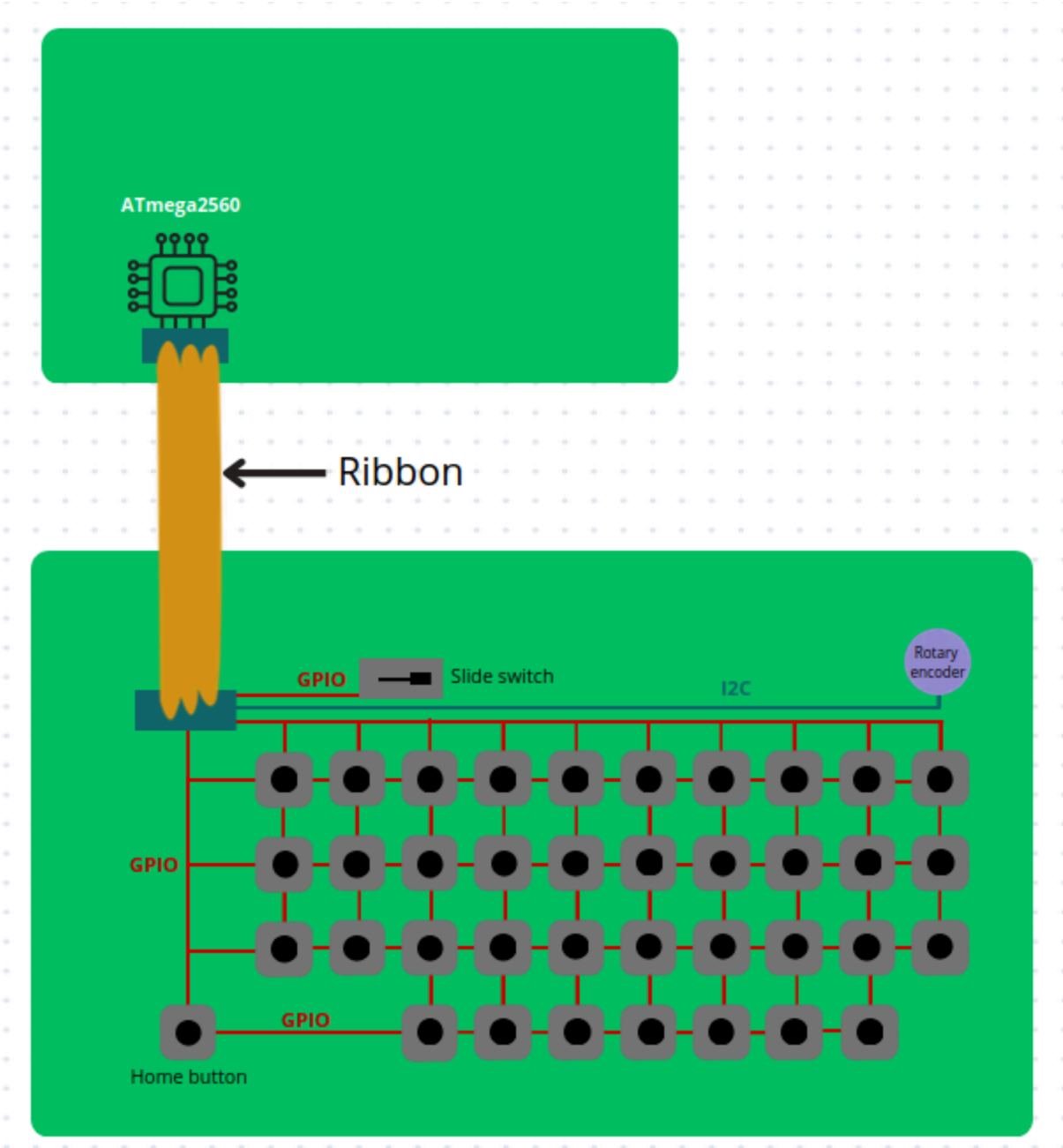
General



Keyboard


ATmega2560
GPIO
Slide switch
I2C
Rotary encoder
GPIO
GPIO
Home button

PCB organisation



IV. Bill of Materials

Qty	Component	Description	URL	Total price	Datasheet
1	ATMega2560-16AU	MCU	LCSC	12,51€	Datasheet
1	ILI9488	Screen LCD SPI	AliExpress	10,49€	Datasheet
2	GC9A01A	Screen LCD SPI rond	AliExpress	1,98€ (0,99€)	Datasheet
1	TYPE-C-31-M-14	USB Type-C	LCSC	1,65$	Datasheet
38	DTS-644R	Pushbutton	AliExpress	1,83€	Datasheet
1	EC11	rotary encoder	AliExpress	1,59€	Datasheet
1	SK-12D07-VG4NSPA	Slide switch	AliExpress	1,71€	Datasheet
3	MTXS0108EPWR	Logic converter	LCSC	1.14€	Datasheet
1	RT9080N-08GJ5	LDO	LCSC	0,60€	Datasheet
1	CJ02-160002020A20	Quartz 16MHz	LCSC	0,25€	Datasheet
1	MIC SD 256mb	SD	AliExpress	1,79€	/
5	KP1608 SGD	Blue LEDs	LCSC	1,44€	Datasheet
1	SML-P11VTT86	Red LEDs	LCSC	0,73€	Datasheet
1	Flat Ribbon	Ribbon 26 pin	AliExpress	3,35€	/
2	Connec femelle IDC	female 26 pin	AliExpress	2.72€	/
2	IDC straight	male 26 pin	AliExpress	2,46€	/
1	CH340C	IC usb - serial	LCSC	0,58€	Datasheet
1	MCP23017	i2c expander	Aliexpress	2,01€	/
1	ESP32 STM32	Sd card reader	aliexpress	1,92€	/
Price of things to consider: the cost of two PCBs.			**TOTAL**	**50,75**€	

If you add the price of the main PCB (~2€), the price of the keyboard PCB (~2€) and the price of the 3D printed case (632 grams, 15,17€), the total is: **69,92€**

V. Pinout

ATmega2560 - Power and quartz

GND	11, 32, 62, 81, 99
VCC	10, 31, 61, 80, 100
XTAL1	34
XTAL2	33

ATmega2560 - Resets

RESET ATmega	30 - RESET
RESET main screen	7 - PE5
RESET eyes	17 - PH5

ATMega2560 - AVR ISP

SCK	20 - PB1
MOSI	21 - PB2
MISO	22 - PB3
RESET	30 - RESET

ATmega2560 - SPI

SS (Slave Select)	19 - PB0
SCK	20 - PB1
MOSI	21 - PB2
MISO	22 - PB3
Data/Command Screens	18 - PH6
Chip select main screen	6 - PE4
Chip select left eye	16 - PH4
Chip select right eye	15 - PH3
Chip select SD	9 - PE7
BLK/LED	8 - PE6

ATmega2560 - UART

RX0	2 - PE0/RXD0
TX0	3 - PE1/TXD0

ATmega2560 - Keyboard rows

C1	57 - PC4
C2	58 - PC5
C3	59 - PC6
C4	60 - PC7
C5	63 - PJ0
C6	64 - PJ1
C7	65 - PJ2
C8	66 - PJ3
C9	67 - PJ4
C10	68 - PJ5

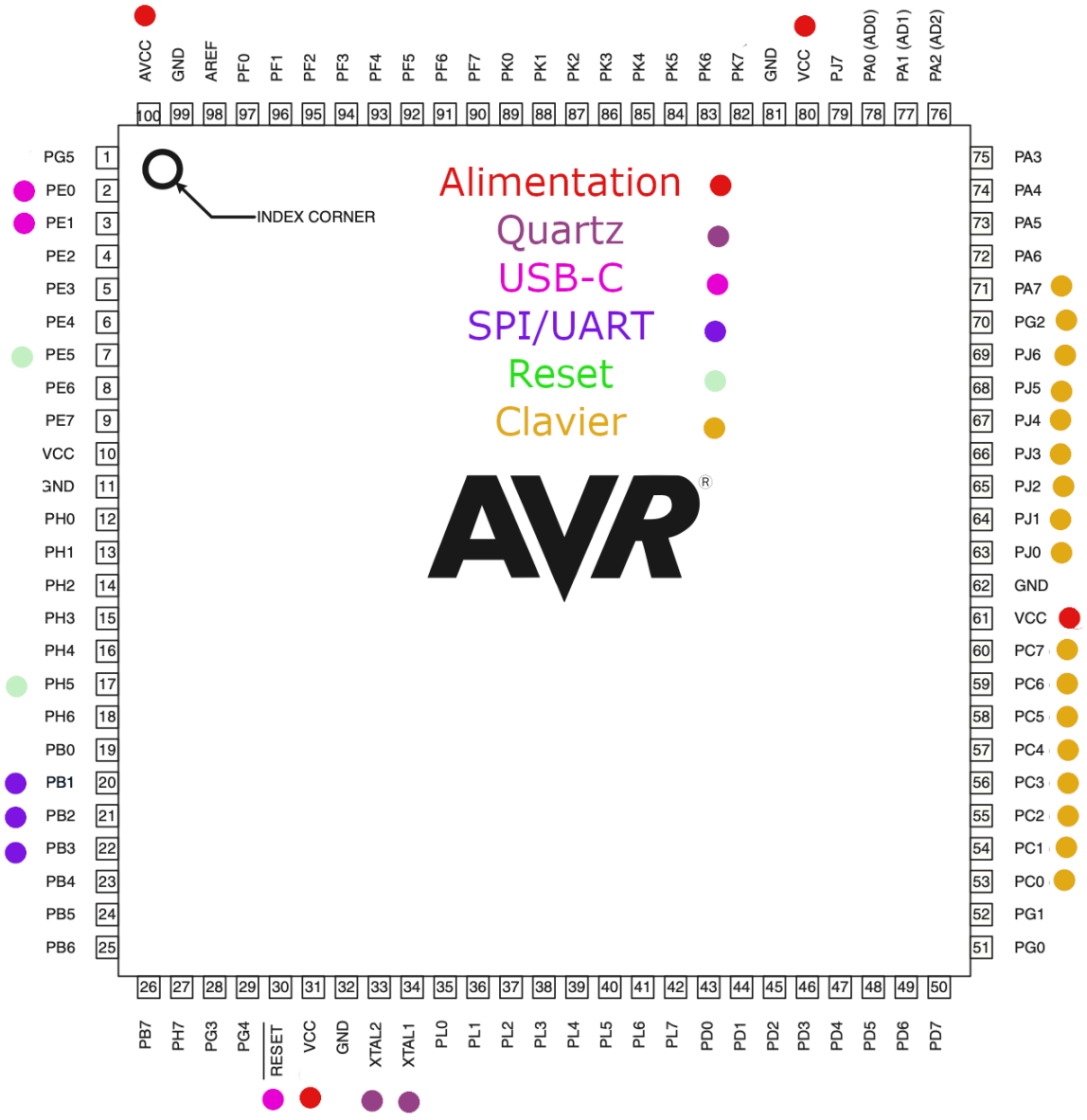
ATmega2560 - Keyboard lines

R1	53 - PC0
R2	54 - PC1
R3	55 - PC2
R4	56 - PC3

ATmega2560 - Rotary encoder and slide switch

i2c expander-SDA	44 - PD1
i2c expander-SCL	43 - PD0
Slide switch 1	69 - PJ6
Slide switch 2	70 - PG2
Slide switch 3	71 - PA7

Pinout : visualisation



VI. Power budget

USB Type-C (Alimentation) : 2A, 5V, 10W

Component	Quantity	Maximum intensity	Voltage	Power
ATMega2560	1	0.06A	5V	0.6W
SPI LCD Screen	1	0.02A	3.3V	0.066W
SPI LCD Screen rounded	2	0.012A	3.3V	0,080W

TOTAL: 0.746W

VII. Protocols

Programming interface

By default, the ATMega2560 uses AVR ISP, we have to make a choice about which programmation interface we are going to pick, so we would have to consider how our USB Type-C connector will transfer data to the MCU and through which communication protocol. As per page 325 of the ATMega2560 datasheet, UART could be used but it's not etched in stone yet for our project.

SPI: Serial Peripheral Interface

Serial Peripheral Interface (SPI) is a bidirectional synchronous communication protocol. It uses 4 lines and leverages a master/slave architecture. The master can pick which slave he wants to talk to through the Slave Select pin (SS).

SLCK : Serial Clock.

MOSI: Master Output, Slave Input.

MISO: Master Input, Slave Output.

SS: Slave Select.

We will use SPI for the communication between the micro-controller and the screens.

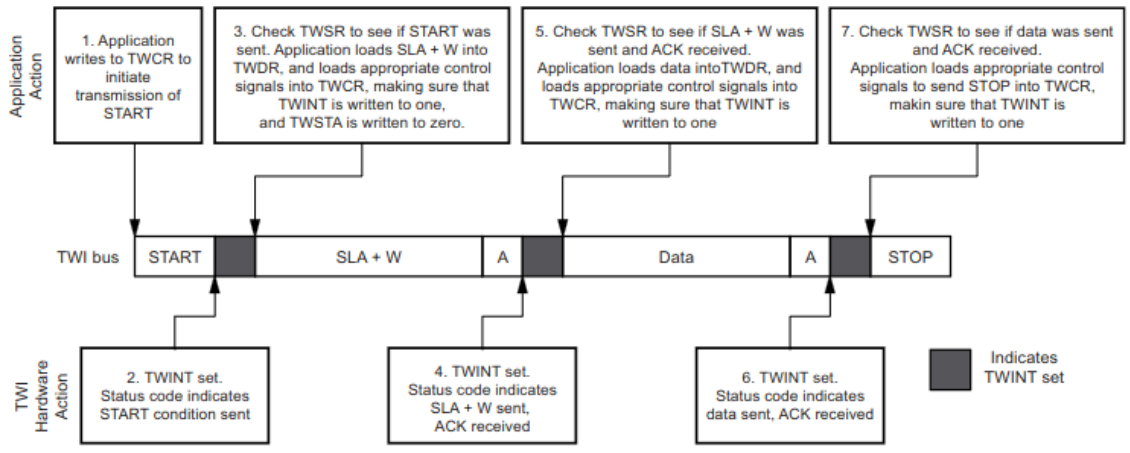
I2C: Inter-Integrated Circuit

Two-Wire Interface, better known as Inter-Integrated Circuits (I^2C or I2C) is a bus connecting several circuits. Each piece of equipment has a role: master or slave. Communication can take place between one master and one slave at least only, and they can solely be initiated by the master. Roles are not constant as a master can become a slave and vice versa. Several masters can coexist on the same I2C line, this process is called multi-master communication.

The two lines necessary for I2C are the following:

- Serial Data Line (SDA): Bidirectional data line.
- Serial Clock Line (SCL): Bidirectional line for the clock used to synchronise communications

Figure 21-10. Interfacing the Application to the TWI in a Typical Transmission



In our project, we will use I2C for the communication between the MCU and the rotary encoder.

UART: Universal Asynchronous Receiver Transmitter

UART is a protocol using asynchronous serial communication. For instance, Device A sends data to Device B bit by bit, with a pre-set baud rate. It should be noted that a synchronous version exists : USART (Universal Synchronous/Asynchronous Receiver Transmitter).

An UART frame is made of the following bits:

- Start bit: Always equal to 0, so the receiver knows there's an inbound communication.
- Data bits (5 to 9 bits): The data is sent in a specific order, from the LSB (Least Significant Bit) to the MSB (Most Significant Bit).
- Parity bit (optional): Either 0 or 1 depending on the convention, it is used to make sure the data hasn't been truncated during the transmission.
- Stop bit: Always equal to 1, signals the end of the transmission.

In our case, we will use UART to help us during debugging as it will allow us to send and receive data on a computer through the `screen` command.

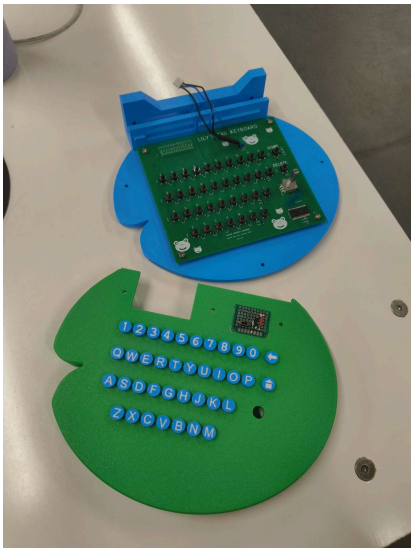
VIII. Bonus

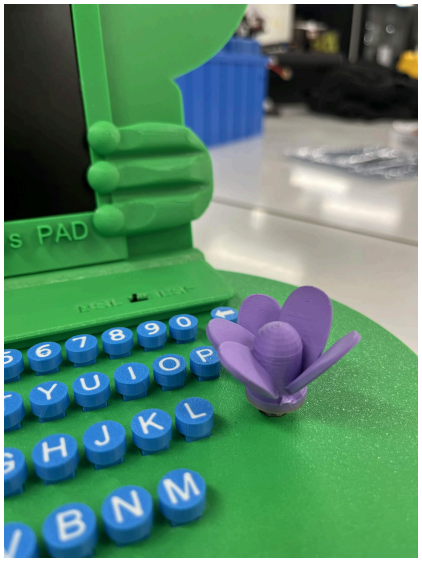
1 - Enclosure design and integration.

first mock-up.

first prototype and v2.

v2 with keyboard.



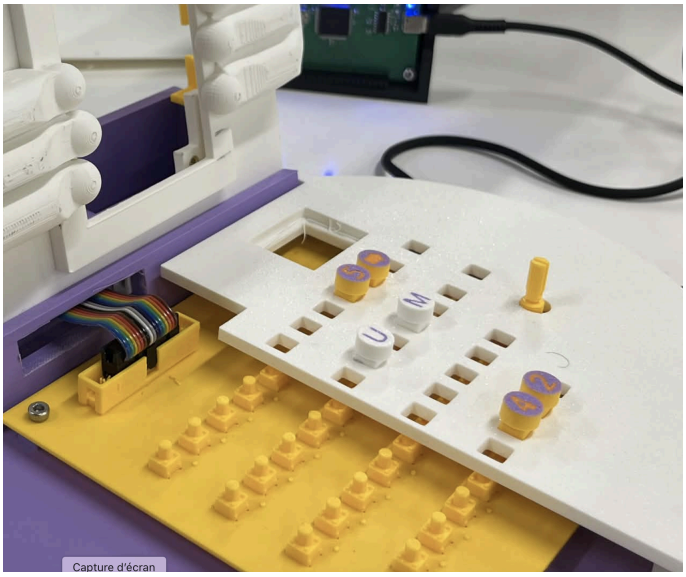


Final version.

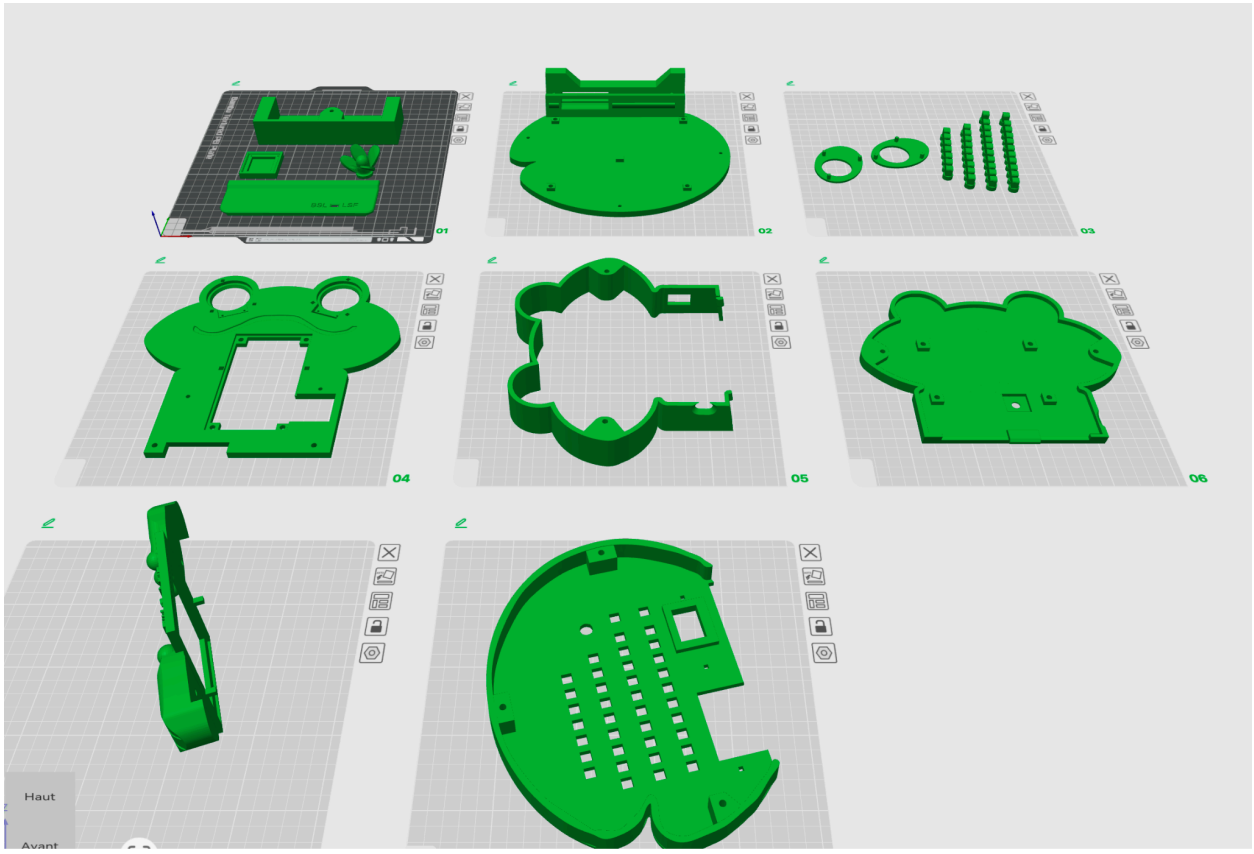
Starting with the idea of making an attractive frog upon a lilyPad. We developed the idea around the PCB's and screens. The casing was designed in Fusion by autodesk and exported in STL file format and then 3D printed.

In order to gain time in the design process, we also 3D printed the pcb's, allowing us to make the casing to an acceptable tolerance level, notably for the keyboard keys.

Thus the case was made before the pcb was physically assembled. It took 25 hours to print.



The cost of printing one complete case is 15,18 euros at current prices.


01
02
03
04
05
06
Haut
Avant

2 - Low-power modes and battery optimisation.

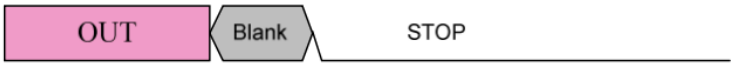
Ideas:

- Sleep Mode
- Dimming of the screen after a delay (ex 15s) then automatic shutdown of the screen and switching of the MCU to low consommation when not in use for a certain time (30s)
- => being able to show that it consumes less?

The ATMega2560 has several power options[1].

What we did:

- Sleep mode, close the frog eyes and turn off the backlight of the main screen after ~20sec.

Description	This command will cause the ILI9488 enter the minimum power consumption mode. In this mode, the DC/DC converter, Internal oscillator, and panel scanning are all stopped.  The MCU interface and memory are still working, and the memory keeps its contents. Dimming function does not work when the mode changes from Sleep OUT to Sleep IN. X = void

ILI9488 datasheet p.165

-

3 - Automated testing and continuous integration of firmware.

Continuous integration: github actions

https://docs.github.com/fr/actions
=> allows you to set an action, such as push, and it will execute code like automated tests.

-> we can make it so that it doesn't push/merge if it doesn't pass the tests, for example

https://github.com/xanderhendriks/rpi-embedded-target-action-runner

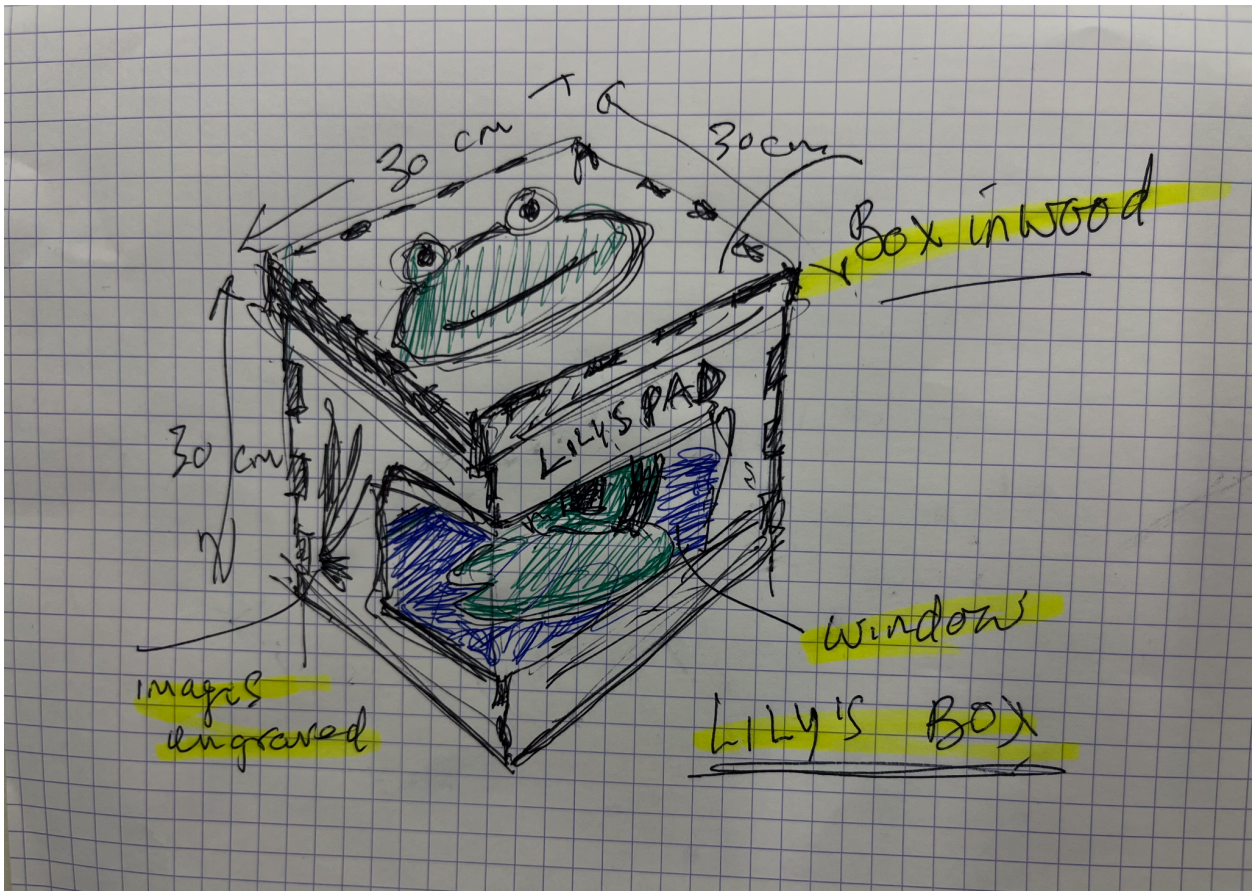
[1] ATmega640/V-1280/V-1281/V-2560/V-2561/V Datasheet, p.50

4 - User-friendly documentation

user-friendly-documentation-lily.pdf

5 - What would we change in a V2 of lily's PAD.

Packaging research



This is an non-exhaustive list but only a few pointers as to what we would add to a new version 2 of such a project.

A serious market study:

1. Possible selling markets, School, Families etc..
2. Pricing with a 3 year plan of selling targets, In France alone there around 700 children born deaf every year. So a realistic plan could be a growth of 100 units per year.

Technical:

1. Add rechargeable batteries (by USB) to make fully autonomous
2. Add better and faster MCU enabling faster processing of images
3. Add better LCD so the display can show animated files (hand movements etc)
4. Add flash RAM chip (NAND/NOR memory) so we can do away with the volatile SD reader card.
5. Add a small work dictionary of signs like Dad, Mum, i'm hungry etc.
6. Add a simple way to update, or upgrade firmware via Github using a small app.

Aesthetical changes:

1. Make different casings that could appeal to different clients
2. Develop the casing as to be foldable or detachable (screen / keyboard) for ease of packaging and delivery.
3. Packaging, a target should be seductive but non costly
4. Add different graphical themes available within Lily'sPAD

Administration:

1. Adapt the project as to have a certification CE
2. Register a model and brand with the INPI. (protection against plagiarism)
3. Approach investors ready to become financial partners on such an adventure

IX. Glossary

GPIO: General Purpose Input/Output

GPIO pins can be used for either input or output purposes, they are not committed to a specific task, thus they are pretty versatile. Using a microcontroller's registers, one can set them to read signals or write them using the dedicated registers.

In our project, we will use GPIO pins for the keyboard keys

Change logs

Changelog-CDC-lily.pdf